Workout Plan

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find

Domain: Integer Linear Programming

Universe

- Schedule: \{Mon, Tue, Wed, Thu, Fri\}

Variables

Symbol	Meaning	Type / Range
x_{ij}	assignment decision	\{0, 1\}
n	number of binary variables = 50	\mathbb{{Z}}^+

Structure

Design a 5-day weekly workout plan selecting exercises from a catalog of 10 exercises.
Each exercise targets one or more muscle groups (chest, back, legs, shoulders, arms, core)
and has a known duration and set count. The plan must:

1. Meet minimum weekly volume targets (total sets) for each of the six muscle groups.
2. Stay within a 60-minute time limit per session.
3. Never schedule the same exercise on two consecutive days.
4. Minimize total weekly workout time.

Real-World Mapping

Real-World Concept	Mathematical Object
schedule	x: solution vector (schedule)

Constraints

(1) Integer Linear Programming *optimizing a linear objective subject to linea*

(2) Muscle volume *total weekly sets targeting a muscle group,*

(3) Consecutive-day constraint *for each exercise, the sum of its binary vari*

(4)	Session time budget	*per-day sum of exercise durations must not*
(5)	Independent verification	*a separate `verify()` function re-checks eve*

Approach

Suggested approach: ILP (PuLP/CBC, Branch & Bound)

Complexity class:

Available tools: ILP

2. Solution

Answer:	5-element schedule: Mon -> [Bench Press], Tue -> [Pull-Up, Plank], Wed -:
Optimal:	Yes
Feasible:	Yes
Algorithm:	ILP (PuLP/CBC, Branch & Bound)
Time:	0.0347s
Certificate:	Branch-and-bound solved to optimality.

Solution Details

Schedule:
  Mon -> Bench Press
  Tue -> Pull-Up, Plank
  Wed -> Romanian Deadlift
  Thu -> Bench Press, Plank
  Fri -> Incline Dumbbell Press, Pull-Up, Squat, Romanian Deadlift

Method: Integer Linear Programming (ILP) via PuLP/CBC.

- 50 binary variables (10 exercises x 5 days)
- Constraints:
  - 6 muscle volume constraints (one per muscle group, >= target)
  - 5 session time-limit constraints (one per day, <= 60 min)
  - 40 consecutive-day constraints (10 exercises x 4 consecutive-day pairs)
- Objective: minimize sum(duration[e] * x[e, d]) over all (exercise, day) pairs
- Solved via CBC Branch & Bound (exact optimal for this small instance)

Verification

✓	All Exercises Valid	Passed
✓	Time Limits Respected	Passed
✓	Muscle Targets Met	Passed
✓	No Consecutive Repeats	Passed
✓	No Intraday Duplicates	Passed
✓	Feasibility	All constraints satisfied
✓	Optimality	ILP (PuLP/CBC, Branch & Bound)

3. Interpretation

The Question

Design a 5-day weekly workout plan selecting exercises from a catalog of 10 exercises.

The Answer

A optimal solution was found.

What This Means

- Optimal 5-day exercise schedule with per-day breakdown
- Total weekly workout time (minimized)
- Weekly muscle volume vs. targets (all PASS)
- Daily time-usage bars (all within 60-minute cap)
- Exercise frequency table
- Independent verification of all six constraint checks (all PASS)

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Recommendations

1. The solution is provably optimal; no further improvement is possible within this model.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.
- Solutio